

UNIVERSITY OF CALGARY
FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS & STATISTICS

FINAL EXAMINATION
LINEAR METHODS I – MATH 211
ALL SECTIONS – WINTER 2017

Date: April 24, 2017

Time: 3 hours

SECTION NUMBER	STUDENT I.D. NUMBER	FIRST NAME	LAST NAME

EXAMINATION RULES

1. This is a closed book examination.
2. No aids are allowed for this examination.
3. Provide all answers on the scantron sheet.
4. Scantron sheets must be filled in during the exam time period. No additional time will be granted to fill in the scantron form.
5. The use of personal electronic or communication devices is prohibited.
6. A University of Calgary Student ID card is required to write the Final Examination and could be requested for midterm examinations. If adequate ID isn't presented, the student must complete an Identification Form.
7. Students late in arriving will not be permitted after one-half hour of the examination time has passed.
8. No student will be permitted to leave the examination room during the first 30 minutes, nor during the last 15 minutes of the examination. Students must stop writing and hand in their exam immediately when time expires.
9. All inquiries and requests must be addressed to the exam supervisor.
10. Students are strictly cautioned against:
 - (a) communicating to other students;
 - (b) leaving answer papers exposed to view;
 - (c) attempting to read other students' examination papers.
11. During the final examination, if a student becomes ill or receives word of domestic affliction, the student must report to the Invigilator, hand in the unfinished paper and request that it be cancelled. If ill, the student must report immediately to a physician/counselor for a medical note to support a deferred examination application.
12. Once the examination has been handed in for marking, a student cannot request that the examination be cancelled. Retroactive withdrawals from the course will be denied.
13. Failure to comply with these regulations will result in rejection of the examination paper.

Part A: Mark your answers on the scantron sheet provided.

Each question is worth 2 points.

1. Consider the following augmented matrix in which $*$ denotes arbitrary numbers and $\blacksquare$ denotes **nonzero** numbers.

$$\left[\begin{array}{cccc|c} \blacksquare & * & * & * & * \\ 0 & * & \blacksquare & * & * \\ 0 & 0 & 0 & * & \blacksquare \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

What can you say about the corresponding system of linear equations?

- (a) It has no solution.
 - (b) It has a unique solution.
 - (c) The general solution has exactly one parameter.
 - (d) The general solution has exactly two parameters.
 - (e) There is insufficient information to determine the answer.
2. Find all values of a and b for which the system

$$\begin{array}{rrcrcl} 2x & + & 3y & + & z & = & b \\ 6x & + & a^2y & + & (b^2 - 1)z & = & 2a \end{array}$$

has infinitely many solutions.

- (a) $a = \pm 3$ and $b = \pm 2$.
 - (b) $a \neq \pm 3$ and $b \neq \pm 2$.
 - (c) $a = 3$ and $b = 2$, and $a = -3$ and $b = -2$.
 - (d) $a = 3$ and $b = -2$, and $a = -3$ and $b = 2$.
 - (e) None of the above.
3. Consider the matrix

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}.$$

What can be said about the equation $AY = BA$, where both Y and B are 2×2 matrices?

- (a) There are no Y satisfying the equation, no matter the matrix B .
- (b) There is a unique Y satisfying the equation, no matter the matrix B .
- (c) There are infinitely many Y satisfying the equation, no matter the matrix B .
- (d) For some matrix B , there are infinitely many Y satisfying the equation.
- (e) None of the above.

4. Let A and B be $n \times n$ matrices.

Which of the following statements is ALWAYS true?

- (a) $(A + B)^{-1} = A^{-1} + B^{-1}$.
- (b) $(AB)^{-1} = A^{-1}B^{-1}$.
- (c) $\det(A + B^T) = \det(A^T + B)$.
- (d) If $AB = 0$, then $BA = 0$.
- (e) $AB - BA = 0$.

5. Let $A = \begin{bmatrix} 1 & 0 & -6 \\ 4 & 2 & -5 \\ 0 & 0 & -2 \end{bmatrix}$.

Then $\det(\operatorname{adj} A)$ is equal to:

- (a) -4 .
- (b) 16 .
- (c) -16 .
- (d) -64 .
- (e) None of the above.

6. Find all values of x such that the matrix $A = \begin{bmatrix} x-1 & 1 \\ -1 & x-1 \end{bmatrix}$ is invertible.

- (a) All values of x .
- (b) For $x = 1 + i$ and $x = \overline{1 + i}$.
- (c) For $x \neq 1 + i$.
- (d) For $x \neq 1 \pm i$.
- (e) None of the above.

7. The shortest distance from the plane T_1 with equation $-x + y - 2z = -7$ to the plane T_2 with equation $x - y + 2z = 1$ is:

(a) 0.
(b) 6.
(c) $\sqrt{6}$.
(d) $1/\sqrt{6}$.
(e) -6 .
(f) None of the above.

8. The shortest distance between the following two lines in $\mathbb{R}^3$

$$L_1 : \begin{cases} x = 2 - t \\ y = 2 + t \\ z = -1 - 2t \end{cases} \qquad L_2 : \begin{cases} x = 1 + 2s \\ y = 1 - 3s \\ z = 1 + 3s \end{cases}$$

is:

(a) 0.
(b) 6.
(c) $6/11$.
(d) $6/\sqrt{11}$.
(e) None of the above.

9. What is the length of the cross product of two vectors of length 2 and included angle $\pi/4$?

(a) 2.
(b) $2\sqrt{2}$.
(c) $-2\sqrt{2}$.
(d) $4\sqrt{2}$.
(e) None of the above.

10. Consider the following four points $\mathbb{R}^3$: $A(1, 0, 0)$, $B(1, 1, 1)$, $C(0, 0, 1)$, and $D(1, 0, -1)$.

The plane containing all four points has equation:

- (a) $2x - y + z = 2$.
- (b) $3x - y + z = 3$.
- (c) $x - y + z = 1$.
- (d) There is no such plane.
- (e) There is such a plane, but none of the above equations are correct.

11. Consider the following two vectors: $\vec{u} = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ -4 \\ -4 \end{bmatrix}$.

Which of the following statements is true about these vectors ?

- (a) They are parallel.
- (b) They are orthogonal.
- (c) They are neither parallel nor orthogonal, but their included angle is acute.
- (d) They are neither parallel nor orthogonal, but their included angle is obtuse.
- (e) None of the above.

12. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given such that

$$T\left(\begin{bmatrix} 2 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} a \\ b \end{bmatrix} \text{ and } T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} c \\ d \end{bmatrix}.$$

Then the matrix of T is:

- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- (b) $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$.
- (c) $\begin{bmatrix} a - c & a - c \\ b - d & b - d \end{bmatrix}$.
- (d) $\begin{bmatrix} a - c & a - 2c \\ b - d & b - 2d \end{bmatrix}$.
- (e) None of the above.

13. Let $Q_{-1} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation which is the reflection in the line $y = -x$.

Then which of the following equals $Q_{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$?

- (a) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (b) $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ (c) $\begin{bmatrix} -1 \\ -2 \end{bmatrix}$ (d) $\begin{bmatrix} -2 \\ -1 \end{bmatrix}$ (e) None of the above.

14. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation such that

$$T \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{and} \quad T \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Find $T^{-1} \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} \right)$.

- (a) $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (d) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ (e) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

15. Let $z = 1 + 2i$ and $w = 2 - i$ be complex numbers. Then $\frac{\overline{w - z}}{\bar{z}}$ equals:

- (a) $(6 - i)/5$.
(b) $-1 + i$.
(c) $1 + i$.
(d) $-1 - i$.
(e) None of the above.

16. Suppose $z = a + bi$ is a root of a real quadratic polynomial.

Which of the following statements is ALWAYS true?

- (a) $\bar{z}$ is also a root.
- (b) $-z$ is also a root.
- (c) z is a root of multiplicity 2.
- (d) $z + \bar{z}$ is also a root.
- (e) None of the above is always true.

17. If $z = -1 - i$, then z^4 equals:

- (a) 4
- (b) $\sqrt{2}e^{5\pi i/4}$
- (c) $4e^{5\pi i}$
- (d) $4e^{5\pi i/4}$
- (e) None of the above.

18. If $z^4 = -1 + \sqrt{3}i$, then the argument of one of the roots z is:

- (a) $\pi/3$.
- (b) $2\pi/3$.
- (c) $3\pi/3$.
- (d) $4\pi/3$.
- (e) None of the above.

19. Consider an $n \times n$ matrix A . Which of the following statements is ALWAYS true?

- (a) A is always diagonalizable.
- (b) If the general solution of $(xI - A)X = 0$ has infinitely many solutions, then A is diagonalizable.
- (c) If the characteristic polynomial of A has a single root of multiplicity n , then A is diagonalizable.
- (d) If the characteristic polynomial of A has n distinct roots, then A is diagonalizable.
- (e) None of the above.

20. The eigenvalues of $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ -2 & -2 & -1 \end{bmatrix}$ are:

- (a) 1, -1 , and 3.
- (b) 1, $1 - 2i$ and $1 + 2i$.
- (c) 3 and -1 , with -1 having multiplicity two.
- (d) 1 and -1 , with 1 having multiplicity two.
- (e) None of the above.

21. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Which of the following statements is true?

- (a) $A^2 = A$.
- (b) A is not diagonalizable.
- (c) A is not invertible.
- (d) $\text{adj } A = A$.
- (e) All of the above statements are false.

22. Consider the Markov Chain $S_{n+1} = PS_n$ where the transition matrix is given by:

$$P = \begin{bmatrix} 1/3 & 1/4 \\ 2/3 & 3/4 \end{bmatrix}.$$

Which of the following is a steady state vector of this Markov Chain?

(a) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}.$

(b) $\begin{bmatrix} 1/3 \\ 2/3 \end{bmatrix}.$

(c) $\begin{bmatrix} 3/11 \\ 8/11 \end{bmatrix}.$

(d) $\begin{bmatrix} 3/8 \\ 5/8 \end{bmatrix}.$

(e) None of the above is a steady state vector for the Markov Chain.

23. Consider the Linear Dynamical System $V_{k+1} = AV_k$ where

$$A = \begin{bmatrix} -2 & 0 \\ 3 & 1 \end{bmatrix}, \text{ and } V_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Which of the following statement is correct?

(a) $\lambda = -1$ is a dominant eigenvalue of A .

(b) $\lambda = +1$ is a dominant eigenvalue of A .

(c) $\lambda = -2$ is a dominant eigenvalue of A .

(d) $\lambda = +2$ is a dominant eigenvalue of A .

(e) None of the above.

24. Consider the matrix

$$A = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}.$$

Which of the following matrices P diagonalizes A ?

(a) $P = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}.$

(b) $P = \begin{bmatrix} -1 & -1 \\ -3 & 1 \end{bmatrix}.$

(c) $P = \begin{bmatrix} 1 & 3 \\ -1 & -1 \end{bmatrix}.$

(d) $P = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}.$

(e) None of the above.

25. The matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1/3 & 2/3 \end{bmatrix}.$$

It turns out that A diagonalizable, that is $P^{-1}AP = D$ is diagonal, where

$$P = \begin{bmatrix} 1 & -3 \\ 1 & 1 \end{bmatrix} \text{ and } D = \begin{bmatrix} 1 & 0 \\ 0 & -1/3 \end{bmatrix}.$$

For large values of n , the entries of the matrix power A^n are closest to the entries of which of the following matrices?

(a) $\begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}.$

(b) $\begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}.$

(c) $\frac{1}{4} \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}.$

(d) $\frac{1}{4} \begin{bmatrix} 1 & 3 \\ 1 & 3 \end{bmatrix}.$

(e) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$